

Bike Marketplace

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Prepared By	Role
Shefa Akter	Project Master
Sabrina Yasmin Srabony	Team Member

Submitted To: Abdullah Al Shiam

Assistant Professor

Computer Science and Engineering, Netrokona University

Table of Contents TABLE OF CONTENTS

CHAPTER 1: SOFTWARE REQUIREMENTS AND SPECIFICATIONS	
.....	3

1.1 Introduction	
.....	3

1.1.1 Purpose	
.....	3

1.1.2 Document Conventions	
.....	3

1.1.3 Intended Audience and Reading Suggestions	
.....	3

1.1.4 Product Scope	
.....	3

1.1.5 Product Perspective	
.....	4

1.1.6 Product Functions Performed	
.....	4

1.2 User Classes and Characteristics

..... 4

1.2.1 Admin

..... 4

1.2.2 Buyer

..... 5

1.2.3 Seller / Showroom Owner

..... 5

1.3 Operating Environment

..... 5

1.3.1 Software Requirements

..... 5

1.3.2 Hardware Requirements

..... 5

1.3.3 Design and Implementation Constraints

..... 5

1.3.4 User Documentation

..... 6

1.3.5 Assumptions and Dependencies

..... 6

1.4 Data Flow Diagram	6
.....	
1.5 Data Dictionary	10
.....	
1.6 System Features	11
.....	
1.6.1 Buyer	11
.....	
1.6.2 Seller / Showroom Owner	15
.....	
1.6.3 Admin	16
.....	
1.7 External Interface Requirements	18
.....	
1.7.1 User Interfaces	18
.....	

1.7.2 Hardware Interfaces	18
1.7.3 Software Interfaces	18
1.7.4 Communications Interfaces	19
1.8 Other Nonfunctional Requirements	19
1.8.1 Performance Requirements	19
1.8.2 Safety Requirements	19
1.8.3 Software Quality Attributes	19
1.8.4 Security Requirements	20

CHAPTER 1

SOFTWARE REQUIREMENTS AND SPECIFICATIONS

1.1 Introduction

1.1.1 Purpose

This software will help a bike marketplace platform operate more efficiently by digitizing the process of listing, discovering, comparing, and purchasing both brand-new and secondhand motorcycles. It enables buyers to search and compare bikes, sellers to manage their listings, and admins to view and moderate all activity on the platform if required. The purpose of this project is to computerize all details regarding bike listings, seller information, buyer inquiries, and transactions, while adding AI-assisted tools such as fair-price suggestion and a chatbot advisor to support better buying decisions.

1.1.2 Document Conventions

This document features some terminology which readers may be unfamiliar with:

- SRS: Software Requirement Specification
- Front End: This stands for the web pages (HTML/CSS/Bootstrap/JavaScript) that a user will see when he/she accesses the application through a browser.
- Listing: A single bike advertisement created by a Seller, describing one motorcycle for sale.
- Operator/Admin: A user who is authorized to moderate listings, manage users, and maintain platform data.
- Entry: A data unit stored to the database.
- DESC: Description
- PRE: Pre-Condition
- DEP: Dependency
- Sign Up: Creating a new user account.
- Log In: Logging in as an existing user.
- Trusted Seller: A seller whose NID and ownership papers have been verified and approved by the Admin.
- Token Money: A small advance/booking amount paid by a Buyer to reserve a listing, which the buyer can pay either through the payment gateway or directly to the seller if they wish.

1.1.3 Intended Audience and Reading Suggestions

The intended readers of this document are the current and future developers working on the “BIKE MARKETPLACE PROJECT”, and the project supervisor/evaluator.

1.1.4 Product Scope

The system will be used as an application that serves bike buyers, individual sellers, and authorized showrooms. The intention of the system is to increase the number of buyers and sellers who can transact

safely and easily in one place, replacing the current scattered, informal process of buying and selling motorcycles through classifieds, social media groups, or word-of-mouth.

If bike buying/selling is handled informally (through scattered classifieds and social groups), buyers and sellers have to put in considerable effort verifying authenticity, comparing prices across sources, and coordinating communication. Listings can be easily lost, duplicated, or misrepresented, and there is no reliable way to verify a seller's identity or a bike's fair price.

1.1.5 Product Perspective

This Bike Marketplace system is a self-contained system that manages the activities of an online bike marketplace.

Due to improperly managed, scattered listings, buyers currently face a lot of difficulty finding trustworthy sellers and fair prices, while sellers struggle to reach genuine buyers. The fully functional, automated marketplace system developed through this project will eliminate the disadvantages caused by the current informal process by improving reliability, efficiency, and trust. The usage of a database to store user, listing, and transaction details will accommodate easy access, retrieval, search, and manipulation of data. The access limitations provided through role-based access levels will enhance the security of the system. The system will facilitate concurrent access and convenient management of activities for buyers, sellers, and admins alike.

1.1.6 Product Functions Performed

- Provide access to registered users only, based on their role (Buyer, Seller, Admin)
- Registration of new buyers and sellers
- Enable a Seller to create, update, and manage bike listings (new and secondhand)
- Enable a Buyer to search, filter, sort, and compare bike listings
- Provide an AI Smart Pricing Assistant to suggest a fair price for secondhand bikes
- Provide a Price Depreciation Predictor for new bikes
- Provide an EMI/Loan Calculator for prospective buyers
- Provide a Smart AI Chatbot (Bike Advisor) to recommend bikes based on budget/preference
- Provide a Visual Search tool to find visually similar listings from an uploaded photo
- Enable a Buyer to save favourite listings (Wishlist) and saved searches with alerts
- Enable direct Buyer–Seller inquiry and in-app messaging
- Enable advance/token booking and test ride scheduling
- Enable a Buyer to rate and review a Seller after a transaction
- Enable a Seller to apply for Verified/Trusted Seller status
- Admin access to moderate listings, manage users, and review platform analytics
- Automatic flagging of suspicious/fraudulent listings for Admin review

1.2 User Classes and Characteristics

1.2.1 Admin

Admin has full access to the system, which means he/she is able to manage any activity with regard to the system. He/she is the highest-privileged user who can access the system.

Key functions:

- Access, approve, edit, or block any bike listing
- Verify sellers and grant/revoke the Trusted Seller badge
- Add, suspend, or manage user accounts
- Manage brand and category data
- Review flagged/suspicious listings and view platform analytics

1.2.2 Buyer

Buyers can choose their preferred bike from the options provided and can also compare, save, or inquire about listings, and book a test ride or place a token payment. Buyers have access only to their own account data, wishlist, and inquiries.

Key functions:

- Search, filter, and compare bike listings
- Use the AI Pricing Assistant, EMI Calculator, and Chatbot advisor
- Save listings to a wishlist and set saved-search alerts
- Message sellers, book test rides, and place token payments
- Rate and review a seller after a transaction

1.2.3 Seller / Showroom Owner

Sellers can create and manage their own bike listings, respond to buyer inquiries, and manage their own verification and booking requests. Sellers have access only to their own listings and related records.

Key functions:

- Create, edit, and manage bike listings (new and secondhand)
- Apply for Trusted Seller verification
- Respond to buyer inquiries and messages
- Confirm or reschedule test ride/booking requests

1.3 Operating Environment

1.3.1 Software Requirements

- Windows 10/11, macOS, or Linux operating system (server: Linux-based hosting)
- Python 3.x runtime

- MySQL Server 8.x
- A modern web browser (Chrome, Firefox, Edge, or Safari) on the client side

1.3.2 Hardware Requirements

- Core i5 processor or equivalent (development/server machine)
- 4GB RAM minimum (8GB recommended for development)
- 20GB of hard disk space on terminal/development machines
- Server machine with sufficient storage for the database and uploaded images (scalable cloud storage recommended)

1.3.3 Design and Implementation Constraints

- The system is accessible over the internet via an encrypted HTTPS connection.
- The system is only accessible through the Bike Marketplace website/application.
- The database is password-protected and accessible only through the application layer.
- The system uses efficient queries to keep RAM and processing power usage low.
- Each user has an individual ID and password.
- Only the Administrator can access full platform-wide management functions.

1.3.4 User Documentation

As part of the system itself, user documentation is provided to buyers and sellers, giving an overview of the system. It includes a full description of the product and clear, step-by-step guidance for account registration, listing creation, and using the AI tools. Users get the opportunity to use the system without having any trouble. The documentation includes contact details (support email) for users who need help. Topics are grouped logically (Buyer Guide, Seller Guide, Admin Guide) so users can quickly find the information they are looking for.

1.3.5 Assumptions and Dependencies

- Each user has a valid user ID and password.
- The server needs to be running for the system to function.
- Users log in to the system to access any personal record, wishlist, or inquiry.
- Only the Administrator can permanently delete a listing or user account.
- Sellers are assumed to provide accurate information; document authenticity is confirmed through manual Admin review.

1.4 Data Flow Diagram

Data flow diagrams (also called data flow graphs) are commonly used during problem analysis. Data flow diagrams (DFDs) are quite general and are not limited to problem analysis for software requirements specification. DFDs are very useful in understanding a system and can be effectively used during analysis. A DFD shows the flow of data through a system. It views a system as a function that transforms inputs

into desired outputs. Any complex system will not perform this transformation in a “single step,” and data will typically undergo a series of transformations before it becomes the output. The DFD aims to capture the transformations that take place within a system on the input data so that eventually the output data is produced. The agent that performs the transformation of data from one state to another is called a process (or a bubble). So, a DFD shows the movement of data through the different transformations or processes in the system. The processes are shown by named circles, and data flows are represented by named arrows entering or leaving the bubbles. A rectangle represents a source or sink and is a net originator or consumer of data. It should be pointed out that a DFD is not a flowchart — a DFD represents the flow of data, while a flowchart shows the flow of control. A DFD does not represent procedural information.

Figure 1.1: Context Diagram (Level-0 DFD) — BikeBazaar System

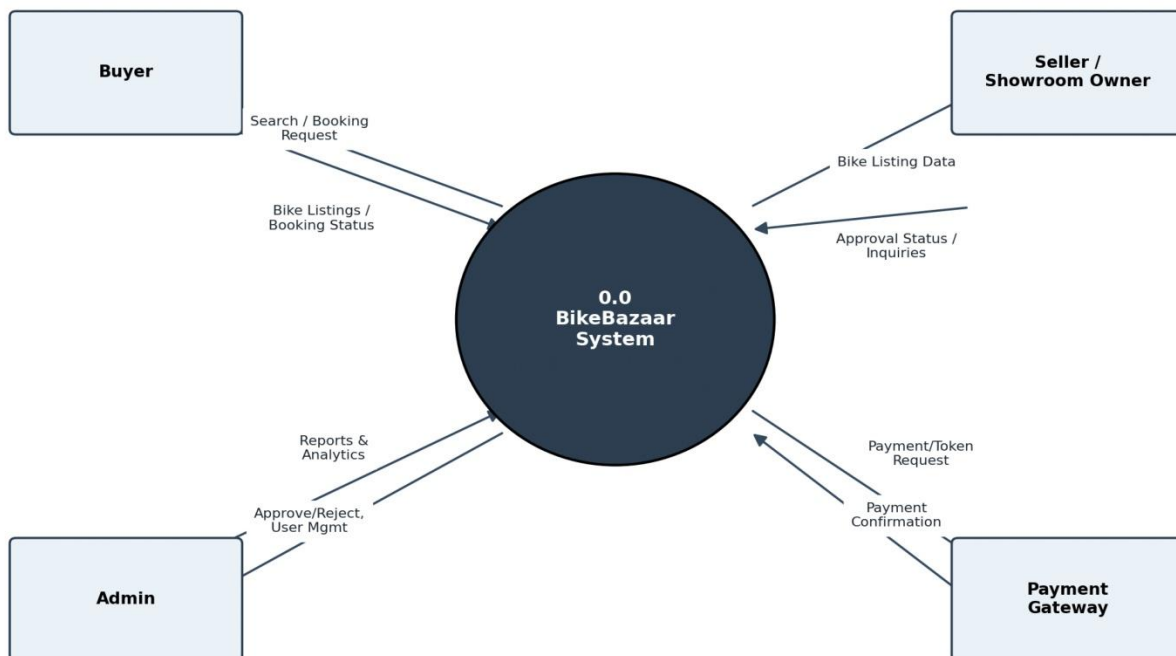


Figure 1.1: Context Diagram (Level-0 DFD) — Bike Marketplace System

As shown in Figure 1.1, the system interacts with four external entities: the Buyer, who submits search and booking requests and receives listing/booking information in return; the Seller/Showroom Owner, who submits bike listing data and receives approval status and inquiries; the Admin, who submits moderation and user-management actions and receives analytics/reports; and the Payment Gateway, which processes token/advance payments and returns payment confirmation.

Figure 1.2: Level-1 Data Flow Diagram — BikeBazaar System

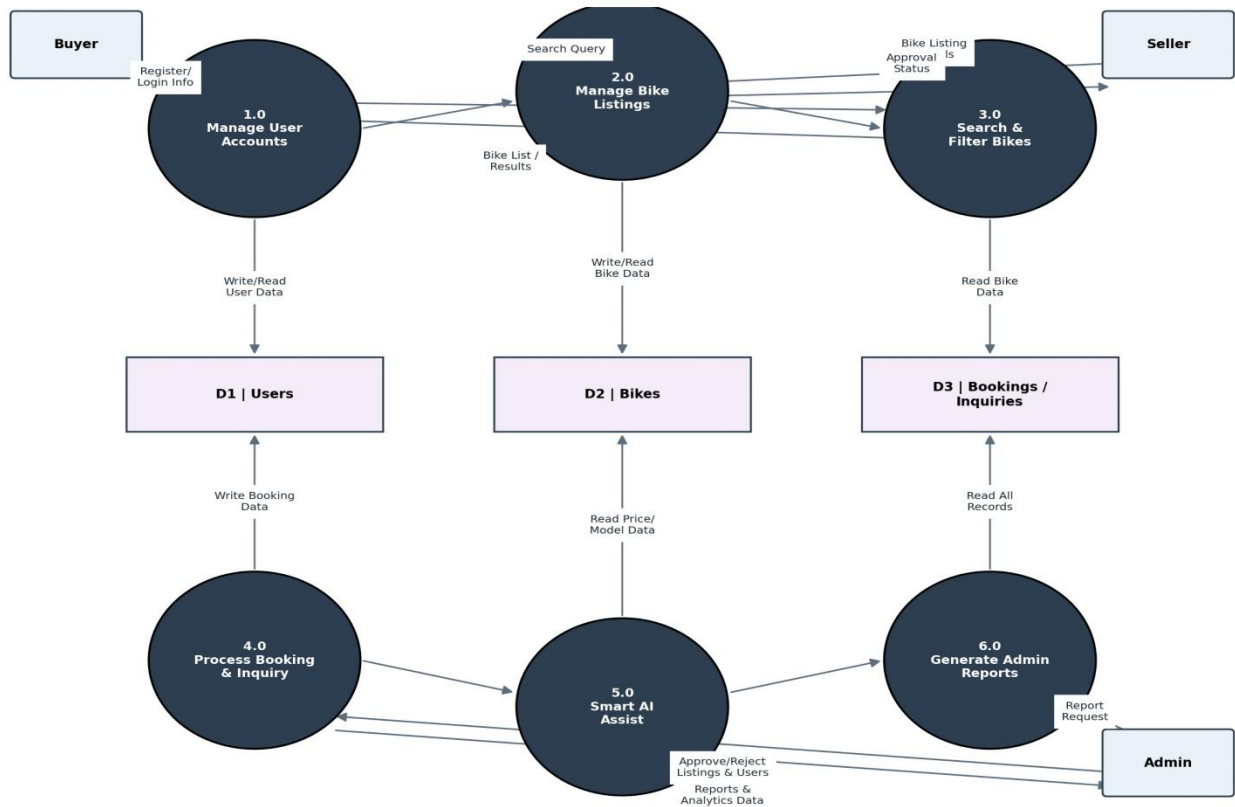


Figure 1.2: Level-1 Data Flow Diagram — Bike Marketplace System Figure 1.2 decomposes the system into six major processes — (1.0) Manage User Accounts, (2.0) Manage Bike Listings, (3.0) Search & Filter Bikes, (4.0) Process Booking & Inquiry, (5.0) Smart AI Assist, and (6.0) Generate Admin Reports — along with three data stores: D1 Users, D2 Bikes, and D3 Bookings/Inquiries.

1.5 Data Dictionary

- Data dictionary is a collection of data that are used as a part of the system.
- It is a record of data about data.

Table name – Users *Primary Key – id*

Field	Data Type	Size	Constraints	Description
id	int	11	Primary Key	Unique ID of the user
name	varchar	100	Not Null	Full name of the user
email	varchar	100	Unique, Not Null	Login email address
password	varchar	255	Not Null	Hashed password
role	enum	-	buyer/seller/admin	Role of the user
phone	varchar	20	-	Contact number
is_verified_seller	boolean	1	Default False	Trusted Seller status

status	enum	-	active/suspended	Account status
--------	------	---	------------------	----------------

Table name – Bikes *Primary Key – id*

Field	Data Type	Size	Constraints	Description
id	int	11	Primary Key	Unique ID of the listing
seller_id	int	11	Foreign Key	References Users.id
title	varchar	150	Not Null	Listing title
brand / model	varchar	100	-	Bike brand and model
manufacturing_year	year	4	-	Year of manufacture
cc	int	5	-	Engine capacity
price	decimal	10,2	Not Null	Asking price (BDT)
condition_type	enum	-	new/used	Listing category
mileage_km	int	10	-	Kilometers run (used bikes)
status	enum	-	available/booked/sold	Listing status
is_approved	boolean	1	Default False	Admin moderation status
view_count	int	10	Default 0	Number of times viewed

Table name – Bike_Images *Primary Key – id*

Field	Data Type	Size	Constraints	Description
id	int	11	Primary Key	Unique ID of the image record
bike_id	int	11	Foreign Key	References Bikes.id
image_path	varchar	255	Not Null	Stored file path of the image

Table name – Inquiries *Primary Key – id*

Field	Data Type	Size	Constraints	Description
id	int	11	Primary Key	Unique ID of the inquiry
buyer_id	int	11	Foreign Key	References Users.id
bike_id	int	11	Foreign Key	References Bikes.id
message	text	-	-	Inquiry message content
status	enum	-	pending/responded/closed	Inquiry status

Table name – Messages *Primary Key – id*

Field	Data Type	Size	Constraints	Description
-------	-----------	------	-------------	-------------

id	int	11	Primary Key	Unique ID of the message
sender_id / receiver_id	int	11	Foreign Key	References Users.id
bike_id	int	11	Foreign Key	References Bikes.id
message	text	-	-	Chat message content
is_read	boolean	1	Default False	Read status of the message

Table name – Reviews *Primary Key – id*

Field	Data Type	Size	Constraints	Description
id	int	11	Primary Key	Unique ID of the review
buyer_id / seller_id	int	11	Foreign Key	References Users.id
rating	int	1	1–5	Star rating given by the buyer
comment	text	-	-	Written review comment

1.6 System Features

This section describes every system feature as a use case, organized by the actor (Buyer, Seller, Admin) who primarily initiates it. Every feature identified for Bike Marketplace — including all AI-assisted tools — is covered below; none has been omitted.

1.6.1 Buyer

REGISTRATION & LOGIN

DESCRIPTION: A new user can register themselves as a Buyer and add their details like name, email, phone number, and password. The buyer entry will be made in the users database.

PRE-CONDITION: The person is a new, unregistered user.

MAIN FLOW OF EVENTS:

1. Buyer selects Sign Up on the login page.
2. A registration form is displayed.
3. Buyer fills in the required details.
4. Buyer submits the form and receives a confirmation.

EXTENSIONS: If necessary fields are left blank, the user is prompted to fill in the missing fields.

POST CONDITION: A new buyer record is added to the Users database, and the buyer can now log in.

SEARCH, FILTER & SORT

DESCRIPTION: The buyer can search the full catalogue of listings and narrow results using a one-click New/Used filter, multi-parameter filters (brand, price range, location, year, CC), and sort options (price, newest, most viewed).

PRE-CONDITION: The catalogue contains at least one approved listing.

MAIN FLOW OF EVENTS:

1. Buyer opens the Browse Listings page.
2. Buyer applies one or more filters and/or a sort order.
3. Matching listings are displayed to the buyer.

EXTENSIONS: If no listing matches the applied filters, the system displays a “no results found” message with a suggestion to broaden the search.

POST CONDITION: The filtered/sorted list of listings is displayed to the buyer.

BIKE COMPARISON

DESCRIPTION: The buyer can select exactly two listings (new or used, in any combination) and view their brand, model, price, year, CC, and mileage side-by-side in a comparison table.

PRE-CONDITION: The buyer has selected two valid, approved listings.

MAIN FLOW OF EVENTS:

1. Buyer selects a first bike and taps “Compare”.
2. Buyer selects a second bike.
3. The system displays both bikes' specifications side-by-side.

EXTENSIONS: If the buyer attempts to compare more than two bikes at once, the system restricts the selection to two.

POST CONDITION: A side-by-side comparison table is displayed to the buyer.

AI SMART PRICING ASSISTANT

DESCRIPTION: The buyer enters (or views, on a secondhand listing) the brand, model, manufacturing year, CC, and mileage, and the system returns an AI-suggested fair market price using a trained regression model, shown alongside the seller's asking price.

PRE-CONDITION: The bike details (brand, model, year, CC, mileage) are available for the listing.

MAIN FLOW OF EVENTS:

1. Buyer opens a secondhand bike's details page.
2. The system sends the bike's attributes to the pricing model.
3. The AI-suggested fair price is displayed next to the seller's asking price.

EXTENSIONS: If the model cannot generate a confident estimate (e.g., a very rare model with insufficient training data), the system displays a message indicating the estimate is unavailable rather than showing an unreliable number.

POST CONDITION: The buyer is shown a data-driven fair-price estimate to support their negotiation/decision.

PRICE DEPRECIATION PREDICTOR

DESCRIPTION: For a new bike listing, the buyer can view an estimated resale-value trend over the next 2–3 years, calculated from a depreciation model.

PRE-CONDITION: The listing is categorized as a Brand New bike.

MAIN FLOW OF EVENTS:

1. Buyer opens a new bike's details page.
2. Buyer views the “Resale Value Trend” graph section.

3. The system displays the estimated depreciation curve for 2–3 years.

EXTENSIONS: If depreciation data for that specific brand/model is unavailable, the system uses a general category-level depreciation estimate and labels it as approximate.

POST CONDITION: The buyer is shown an estimated future resale-value trend for the bike.

EMI / LOAN CALCULATOR

DESCRIPTION: The buyer inputs the bike price, down payment, loan tenure, and interest rate, and the system calculates the estimated monthly EMI and total repayment amount.

PRE-CONDITION: The buyer provides a valid bike price and loan terms.

MAIN FLOW OF EVENTS:

1. Buyer opens the EMI Calculator tool (from a bike's details page or a standalone page).
2. Buyer enters price, down payment, tenure, and interest rate.
3. The system calculates and displays the monthly EMI and total repayment amount.

EXTENSIONS: If the down payment entered is greater than or equal to the bike price, the system displays an EMI of zero with an appropriate message.

POST CONDITION: The buyer receives an estimated monthly installment breakdown.

SMART AI CHATBOT (BIKE ADVISOR)

DESCRIPTION: A 24/7 chatbot asks the buyer for their budget range and condition preference (new/used) and suggests matching listings from the catalogue.

PRE-CONDITION: The buyer opens the chatbot widget, available on any page.

MAIN FLOW OF EVENTS:

1. Buyer opens the chatbot.
2. Chatbot asks for budget range and new/used preference.
3. Buyer responds.
4. Chatbot queries the database and returns matching listings.

EXTENSIONS: If no listing matches the buyer's stated budget/preference, the chatbot suggests the closest available alternatives.

POST CONDITION: The buyer receives a short list of recommended listings from the chatbot.

VISUAL SEARCH

DESCRIPTION: The buyer uploads a photo of a bike, and the system returns a ranked list of visually similar listings already on the platform.

PRE-CONDITION: The buyer uploads a valid image file.

MAIN FLOW OF EVENTS:

1. Buyer opens the Visual Search tool.
2. Buyer uploads a bike photo.
3. The system extracts visual features from the photo and compares them against existing listings.
4. A ranked list of visually similar listings is displayed.

EXTENSIONS: If no sufficiently similar listing is found, the system displays a “no close matches found” message.

POST CONDITION: The buyer is shown a ranked list of visually similar bike listings.

WISHLIST, SAVED SEARCH & PRICE-DROP ALERTS

DESCRIPTION: The buyer can save favourite listings to a wishlist, save a filter combination as a saved search, and receive notifications when new matching listings appear or a wishlisted listing's price drops.

PRE-CONDITION: The buyer is logged in.

MAIN FLOW OF EVENTS:

1. Buyer taps "Add to Wishlist" on a listing, or "Save Search" on a filtered results page.
2. The system stores the wishlist item or saved-search filter against the buyer's account.
3. The system checks new/updated listings against saved searches and wishlist items.
4. A notification is sent to the buyer when a match or price drop occurs.

EXTENSIONS: If the buyer's saved search matches zero new listings within a set period, no notification is sent.

POST CONDITION: The buyer's wishlist/saved search is stored, and relevant notifications are delivered.

INQUIRY & IN-APP MESSAGING

DESCRIPTION: The buyer submits an inquiry linked to a specific listing and can then exchange messages with the seller through an in-app chat thread.

PRE-CONDITION: The buyer is logged in and viewing a valid listing.

MAIN FLOW OF EVENTS:

1. Buyer opens a bike's details page and selects "Contact Seller".
2. Buyer types and sends an inquiry message.
3. The system creates an inquiry record and notifies the seller.
4. Buyer and seller continue the conversation through the in-app messaging thread.

EXTENSIONS: If the seller does not respond within a defined period, the system may send the buyer a reminder that the inquiry is still pending.

POST CONDITION: The inquiry and subsequent messages are stored, and both parties can view the conversation history.

BOOKING, TEST RIDE & TOKEN PAYMENT

DESCRIPTION: The buyer can place an advance/token payment to reserve a listing (which can be paid via the payment gateway or directly to the seller), and/or schedule a test ride with the seller or showroom at an available date and time slot.

MAIN FLOW OF EVENTS:

- Buyer selects "Book Now" or "Schedule Test Ride" on a listing.
- For token payment: buyer completes the payment through the sandbox payment gateway or pays the seller directly.
- For test ride: buyer selects an available date/time slot.
- The system notifies the seller of the booking or test ride request

EXTENSIONS: If the selected time slot becomes unavailable before confirmation, the system prompts the buyer to choose another slot.

POST CONDITION: The listing's status is updated (e.g., to Booked), and the seller receives the booking/test ride request for confirmation.

REVIEW & RATING

DESCRIPTION: After a transaction is marked complete, the buyer can submit a star rating (1–5) and a written comment about the seller.

PRE-CONDITION: The buyer has a transaction marked as complete with that seller.

MAIN FLOW OF EVENTS:

1. Buyer opens the completed transaction/order.
2. Buyer selects a star rating and writes a comment.
3. Buyer submits the review.

EXTENSIONS: A buyer cannot submit more than one review for the same completed transaction.

POST CONDITION: The review is stored and the seller's average rating is updated and displayed on their profile.

1.6.2 Seller / Showroom Owner

REGISTRATION & VERIFIED SELLER APPLICATION

DESCRIPTION: A new user can register as a Seller and, optionally, apply for Trusted Seller verification by uploading a National ID (NID) and the bike's ownership papers.

PRE-CONDITION: The person is a new, unregistered user (or a registered Buyer upgrading to Seller).

MAIN FLOW OF EVENTS:

1. Seller selects Sign Up and completes registration.
2. Seller opens “Apply for Verification” from their dashboard.
3. Seller uploads NID and ownership-paper documents.
4. The request is submitted to the Admin's verification queue.

EXTENSIONS: If a document fails to upload (wrong format/size), the system prompts the seller to re-upload it.

POST CONDITION: The seller account is created, and a pending verification request is created if submitted (approved/rejected later by Admin).

BIKE LISTING MANAGEMENT (NEW & USED)

DESCRIPTION: The seller creates, edits, and manages listings for brand-new or secondhand bikes, including images, category-specific fields (warranty/showroom for new; mileage/papers status for used), and listing status.

PRE-CONDITION: The seller is logged in (and, for showroom listings, ideally verified).

MAIN FLOW OF EVENTS:

1. Seller selects “Add New Listing”.
2. Seller selects condition type (New/Used) and fills in the relevant fields.
3. Seller uploads one or more bike images.
4. Seller submits the listing for Admin approval.

EXTENSIONS: If required fields are left blank, the system prompts the seller to complete them before submission. If the AI Condition Detector (see Admin section) flags an uploaded image as inconsistent with the stated condition, the listing is routed for a closer Admin review before approval.

POST CONDITION: The listing is added to the Bikes database with status “pending approval” until reviewed by the Admin.

RESPOND TO INQUIRIES & MESSAGES

DESCRIPTION: The seller views inquiries received on their listings and responds through the in-app messaging thread.

PRE-CONDITION: The seller has at least one listing with a received inquiry.

MAIN FLOW OF EVENTS:

1. Seller opens the Inquiries/Messages section of their dashboard.
2. Seller selects an inquiry to view the buyer's message.
3. Seller types and sends a reply.

EXTENSIONS: If the seller has multiple pending inquiries, the system lists them in order of most recent first.

POST CONDITION: The reply is added to the conversation thread and the buyer is notified.

MANAGE TEST RIDE / BOOKING REQUESTS

DESCRIPTION: The seller views incoming test ride and token-booking requests and confirms or reschedules them.

PRE-CONDITION: A buyer has submitted a test ride or booking request on the seller's listing.

MAIN FLOW OF EVENTS:

1. Seller opens the Bookings section of their dashboard.
2. Seller reviews the requested date/time or token payment status.
3. Seller confirms, or proposes an alternative time slot.

EXTENSIONS: If the seller does not respond within a defined period, the system may automatically flag the request as needing attention.

POST CONDITION: The booking/test ride status is updated and the buyer is notified of the outcome.

VIEW LISTING ANALYTICS

DESCRIPTION: The seller can view basic performance data for each of their listings, such as the number of times it has been viewed.

PRE-CONDITION: The seller has at least one active listing.

MAIN FLOW OF EVENTS:

1. Seller opens “My Listings” on their dashboard.
2. Seller selects a listing.
3. The system displays the view count and current status for that listing.

EXTENSIONS: If a listing is brand new with zero views, the system displays a view count of zero rather than an error.

POST CONDITION: The seller is shown up-to-date performance data for their listing.

1.6.3 Admin

LISTING MODERATION

DESCRIPTION: The admin reviews newly submitted listings and approves, edits, or blocks them before (or after) they become publicly visible.

PRE-CONDITION: The admin is logged in with valid credentials.

MAIN FLOW OF EVENTS:

1. Admin opens the Listing Moderation queue.
2. Admin reviews a pending listing's details and images.
3. Admin approves, edits, or blocks/rejects the listing.

EXTENSIONS: If a listing is edited by the Admin, the seller is notified of the change.

POST CONDITION: The listing's approval status is updated in the Bikes database, and it becomes publicly visible if approved.

USER MANAGEMENT

DESCRIPTION: The admin can view all registered users and activate or suspend any account found to be in violation of platform rules.

PRE-CONDITION: The admin is logged in with valid credentials.

MAIN FLOW OF EVENTS:

1. Admin opens the User Management section.
2. Admin searches for or selects a specific user.
3. Admin activates, suspends, or otherwise updates the account status.

EXTENSIONS: A suspended user cannot log in until reactivated by an Admin.

POST CONDITION: The user's account status is updated in the Users database.

SELLER VERIFICATION APPROVAL

DESCRIPTION: The admin reviews a seller's submitted NID and ownership-paper documents and approves or rejects the Trusted Seller request.

PRE-CONDITION: A seller has submitted a pending verification request.

MAIN FLOW OF EVENTS:

1. Admin opens the Verification Requests queue.
2. Admin reviews the uploaded NID and ownership documents.
3. Admin approves or rejects the request.

EXTENSIONS: If rejected, the system notifies the seller with a reason and allows them to resubmit corrected documents.

POST CONDITION: The seller's `is_verified_seller` status is updated, and the Trusted Seller badge appears on their profile if approved.

BRAND & CATEGORY MANAGEMENT

DESCRIPTION: The admin maintains the master list of brands and categories used across all listings.

PRE-CONDITION: The admin is logged in with valid credentials.

MAIN FLOW OF EVENTS:

1. Admin opens the Brand/Category Management section.
2. Admin adds, edits, or removes a brand or category entry.

EXTENSIONS: A brand/category already in use by existing listings cannot be deleted; it can only be deactivated.

POST CONDITION: The updated brand/category list is available platform-wide for new listings.

FRAUD / SCAM LISTING REVIEW

DESCRIPTION: The system automatically flags listings that reuse an image already present elsewhere on the platform, or whose asking price is significantly below the AI-suggested fair price, and the admin reviews these flagged listings. This section also covers the AI Condition & Damage Detector, which scans uploaded bike photos and flags a mismatch with the seller's stated condition for the same review queue.

PRE-CONDITION: At least one listing has been auto-flagged, or a buyer has manually reported a listing.

MAIN FLOW OF EVENTS:

1. System auto-flags a listing (duplicate image, abnormal price, or condition mismatch) or a buyer submits a manual report.
2. The flagged listing appears in the Admin's Fraud Review queue.
3. Admin reviews the flag reason and the listing details.
4. Admin blocks the listing, contacts the seller, or dismisses the flag if it is a false positive.

EXTENSIONS: A high volume of false positives on a specific rule (e.g., price threshold) should prompt the Admin/development team to retune that rule's sensitivity.

POST CONDITION: The flagged listing's status is updated (blocked, edited, or cleared), keeping the marketplace trustworthy.

ANALYTICS DASHBOARD

DESCRIPTION: The admin views platform-wide analytics, including total users, total listings, total completed bookings, revenue, and visitor trends.

PRE-CONDITION: The admin is logged in with valid credentials.

MAIN FLOW OF EVENTS:

1. Admin opens the Analytics Dashboard.
2. The system aggregates data from the Users, Bikes, and Inquiries/Bookings tables.
3. Summary statistics and trend graphs are displayed to the Admin.

EXTENSIONS: If a data category has no records yet (e.g., a brand-new platform), the corresponding chart displays a zero/empty state rather than an error.

POST CONDITION: The Admin is shown an up-to-date overview of platform performance.

1.7 External Interface Requirements

1.7.1 User Interfaces

This section provides a description of the inputs into and outputs from the system, along with the hardware, software, and communication interfaces.

- The protocol used is HTTPS.
- The port number used is 443 (HTTPS) with a fallback of 80 (HTTP, redirected to HTTPS).
- The system is addressable via a standard domain name resolving to the server's IPv4 (or IPv6) address.

- Wireframes/prototypes for each core screen (Home, Browse, Bike Details, Comparison, Dashboards) are maintained as a separate UI design artifact.

1.7.2 Hardware Interfaces

Laptop / Desktop / Mobile Device

The purpose of this is to give buyers and sellers access to listings, AI tools, and messaging. To perform such actions smoothly, a reasonably capable device and stable internet connection are needed; otherwise, users may experience delays loading listings or images.

Device Camera

Used optionally by Sellers to capture bike photos for listings and verification documents, and by Buyers for the Visual Search feature.

Wi-Fi / Mobile Data Connection

Required for all client-server communication; used for browsing, uploading images, and real-time messaging.

1.7.3 Software Interfaces

- Python 3.x — Backend programming language.
- Flask / Django — Python web framework for routing and business logic.
- MySQL Server 8.x — Database connectivity and management.
- HTML5, CSS3, Bootstrap 5, JavaScript (ES6) — Frontend structure, styling, and interactivity.
- PyMySQL / SQLAlchemy — Database connector/ORM between Python and MySQL.
- Scikit-learn, Pandas — Used to build and run the AI Smart Pricing model.
- Visual Studio Code / PyCharm — IDE for development.
- Windows 10/11 or Linux — Common development and server operating systems.

1.7.4 Communications Interfaces

- NIC (Network Interface Card) — Allows the server/client machine to connect to a network, wired or wireless.
- TCP/IP protocol — Used to access and share information over the Internet.
- HTTPS/TLS — Encrypts all data transmitted between client and server.
- SMTP — Used for sending email notifications (password reset, booking/inquiry alerts).
- Ethernet / Wi-Fi — Common, low-cost, high-data-rate connectivity for the hosting environment.

1.8 Other Nonfunctional Requirements

1.8.1 Performance Requirements

Response Time: The system will give responses within 2–3 seconds after a search, filter, or AI tool request under normal conditions.

Capacity: The system should support at least 50–100 concurrent users without noticeable degradation, consistent with the scale of an academic/small-scale deployment.

User Interface: The user interface screen will respond within 3 seconds of a user action.

1.8.2 Safety Requirements

If there is extensive damage to a wide portion of the database due to a catastrophic failure, such as a disk crash, the recovery method restores the most recent backup copy of the database from archival storage and reconstructs a more current state by reapplying the operations of committed transactions from the backup log, up to the time of failure. All Admin and Seller operators have unique logins, so the system can identify who is currently logged in; no intruders are allowed, and except for the system Administrator, nobody can alter listing or user data outside their own account.

1.8.3 Software Quality Attributes

- **AVAILABILITY:** The system is available at all times, subject to routine maintenance windows.
- **CORRECTNESS:** A bug-free software that fulfills the correct needs/requirements of the client and end users.
- **MAINTAINABILITY:** The ability to maintain, modify information, and update/fix problems in the system with minimal effort, aided by a modular backend structure.
- **USABILITY:** The software can be used repeatedly by buyers, sellers, and admins without confusion or distortion of results.
- **ACCESSIBILITY:** The Administrator and other users can access the system, but the access level is controlled for each user according to their role.
- **ACCURACY:** Reliability of the information/output — users can depend on and be sure of the outcome (e.g., AI price suggestions, EMI calculations).
- **STABILITY:** The system's output will not change unpredictably over time; the same output will be given for a given input under the same conditions.

1.8.4 Security Requirements

1. All passwords are stored using a strong one-way hashing algorithm (e.g., bcrypt or Werkzeug's password hashing utilities); plain-text passwords are never stored.
2. The system enforces role-based access control (RBAC) so that Buyers, Sellers, and Admins can only access functionality permitted for their role.
3. All user input is validated and sanitized on both client and server sides to prevent SQL injection and Cross-Site Scripting (XSS) attacks.
4. All data transmission between client and server occurs over HTTPS.
5. Uploaded identity documents (NID, ownership papers) are stored securely and are accessible only to Admin accounts for verification purposes.
6. User sessions expire after a defined period of inactivity, requiring re-authentication.